MANASVA DHAKA

Software Engineer | B.Tech Student

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ABOUT ME

B.Tech Information Technology undergraduate at Manipal Institute of Technology with internship experience shipping a production-style full-stack application. Hands-on with REST API design, JWT authentication and role-based access control, MongoDB schema and aggregation design, real-time features with Socket.IO, and automated testing. Strong foundation in Java, C and Python with daily practice in data structures and algorithms, and a growing focus on backend systems and scalable software design. Looking to contribute as a Software Engineering Intern.

EXPERIENCE

Software Development Intern — Adnate IT Solutions Pvt Ltd (Remote) *Jun 2026 – Jul 2026*

- Built a full-stack Student Management System (React, Redux Toolkit, React Router; Node.js, Express, MongoDB) with full CRUD, search, filtering, sorting and pagination.
- Implemented JWT-based authentication with role-based access control (RBAC) and an OTP-driven forgot-password flow with automated email notifications.
- Built authenticated real-time features with Socket.IO, including live user-presence tracking and an admin activity feed that updates on every data change.
- Designed MongoDB aggregation pipelines to power dashboard analytics, including branch-wise distribution, average CGPA and registration trends.
- Added Excel, CSV and PDF export endpoints and implemented audit logging for all administrative actions.
- Wrote unit and integration tests (Jest, Supertest, mongodb-memory-server) covering authentication and core CRUD functionality.

PROJECTS

Distributed File Storage System (In Progress) GitHub *Java · SHA-256 · Streaming I/O*

- Building a GFS/HDFS-inspired storage system in plain Java: files are streamed into 1 MiB chunks, spread round-robin across storage nodes, and reassembled byte-for-byte.
- Verified end-to-end data integrity with SHA-256 checksums; a test confirms corrupted chunks are detected, and stores fail over to the next node on I/O errors.
- Kept memory bounded with fixed-size streaming buffers (64 MiB processed under a 16 MiB heap) and wrote 129 tests plus 5 stress tests, including simulated disk failures.

AI-Powered Customer Support Chat Widget *FastAPI · Python · Gemini API · JavaScript*

- Built an embeddable AI chat widget that answers customer questions strictly from a business's own FAQ content, preventing hallucinated responses.
- Designed a FastAPI backend whose REST endpoint grounds every model response in a configurable knowledge base, keeping API keys server-side only.
- Shipped a dependency-free JavaScript widget that embeds on any website with two script tags.

Personal Portfolio Website *HTML · CSS · JavaScript*

- Responsive portfolio showcasing projects and skills, iterated on UI/UX and accessibility from review feedback.

SKILLS

Languages: Java, C, Python, JavaScript
Web & Backend: React.js, Redux Toolkit, Node.js, Express.js, REST API Design, JWT Authentication & RBAC, Socket.IO
Databases: MongoDB, SQL
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems
Tools & Testing: Git & GitHub, Jest, Supertest

EDUCATION

Manipal Institute of Technology, Bengaluru *2024 – 2028*

Bachelor of Technology (B.Tech), Information Technology • **CGPA: 7.34**

BVB Vidyashram, JLN Marg, Jaipur, Rajasthan

Class 12th: 88.2% | Class 10th: 89%

LANGUAGES

English, Hindi